

The Two-Fugacity Divisor of the Weighted Hénon Reflection Euler Family

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Abstract

The orbit-resolved reflection Euler product of the full Hénon horseshoe depends on a positive defect weight q , but its previously known all-channel continuation was restricted to $q = 1$. We lift each orbit monomial by setting $w = qz$, so that $z^n q^{S_n \chi} = z^{n-S_n \chi} w^{S_n \chi}$. Primitive Möbius subtraction and Euler repetition then give the exact two-variable identity

$$\log \mathcal{Z}^\sharp(z, w) = \sum_{m \geq 1} c_m \frac{2w^m}{1 - z^{2m} - w^{2m}}, \quad c_m = \frac{1}{m} \prod_{\substack{p|m \\ p \text{ odd}}} (1 - p).$$

Every c_m is nonzero. In the bidisk, the hypersurfaces $\mathcal{H}_m = \{z^{2m} + w^{2m} = 1\}$ are smooth and form a locally finite effective divisor, while the channel sum converges normally on its complement. Restriction to $w = qz$, $q > 0$, recovers the weighted orbit product. The m th channel has $2m$ explicitly located roots and a nonzero closed principal coefficient at each root, so every one is exponentially essential. We do not infer a dense natural boundary, introduce a weighted Lind zeta, or construct an operator or arithmetic trace.

1 The weighted packet and the missing global parameter

For the area-preserving map $H(x, y) = (6 - x^2 - y, x)$, the inherited horseshoe coding identifies the odd reflection packets used below with reversal-fixed binary words [2, 1]. Let A_n be the primitive marked packet at odd period n and define

$$\chi(s) = \mathbf{1}\{s_{-1} = s_1\}, \quad S_n \chi(s) = \sum_{j=0}^{n-1} \chi(\sigma^j s).$$

The orbit-resolved product is

$$\mathcal{Z}_{\text{orb}}(z, q) = \prod_{\substack{n \geq 1 \\ n \text{ odd}}} \prod_{\omega \in A_n} \left(1 - z^n q^{S_n \chi(\omega)}\right)^{-1}, \quad q > 0. \quad (1)$$

It is a marked reflection-packet product, not the full Lind zeta of a flip action.

The preceding transfer calculation gives, for $n = 2h + 1$,

$$F_n(q) = 2q(1 + q^2)^h, \quad (2)$$

$$E_n(q) = \sum_{k|n} \mu(k) F_{n/k}(q^k), \quad (3)$$

where F_n counts all reflection words and E_n counts the primitive marked packet with its individual energy weights. If $E(z, q) = \sum_{n \text{ odd}} E_n(q) z^n$, expansion of the Euler factors gives

$$\log \mathcal{Z}_{\text{orb}}(z, q) = \sum_{r \geq 1} \frac{1}{r} E(z^r, q^r) \quad (4)$$

in the disk of absolute convergence. At $q = 1$, grouping the Möbius and repetition indices exposes infinitely many rational channels. Our goal is to retain the full weight without postulating a new external zeta formula.

2 A two-fugacity lift

The inequality $0 \leq S_n \chi \leq n$ permits a bivariate lift of every orbit factor. Set $w = qz$ and write

$$z^n q^{S_n \chi} = z^{n - S_n \chi} w^{S_n \chi}. \quad (5)$$

We denote by $\mathcal{Z}^\sharp(z, w)$ the resulting formal Euler product. Equation (2) immediately becomes a rational all-word function.

Proposition 2.1 (Lifted all-word function). *For sufficiently small z, w ,*

$$F^\sharp(z, w) := \sum_{h \geq 0} 2w(z^2 + w^2)^h = \frac{2w}{1 - z^2 - w^2}. \quad (6)$$

Primitive subtraction and repetition take the forms

$$E^\sharp(z, w) = \sum_{\substack{k \geq 1 \\ k \text{ odd}}} \mu(k) F^\sharp(z^k, w^k), \quad (7)$$

$$\log \mathcal{Z}^\sharp(z, w) = \sum_{r \geq 1} \frac{1}{r} E^\sharp(z^r, w^r). \quad (8)$$

On $w = qz$, these identities agree coefficientwise with (3)–(4).

Proof. In a reflection word of length $2h + 1$, the fixed boundary edge contributes one unit of energy. Each of the remaining h paired edges either changes the symbol, contributing z^2 , or preserves it, contributing w^2 ; the initial binary symbol contributes the factor $2w$. This proves (6). A period- n/k word repeated k times dilates both exponents in (5) by k , so ordinary odd Möbius inversion gives (7). Expanding $-\log(1 - u)$ yields (8). Substitution $w = qz$ reverses (5) and recovers the original formulas. $\square$

3 Exact channel regrouping

For $m \geq 1$, define

$$\Psi_m(z, w) = \frac{2w^m}{1 - z^{2m} - w^{2m}}. \quad (9)$$

Theorem 3.1 (Weighted scalar-channel identity). *In the initial domain of absolute convergence,*

$$\log \mathcal{Z}^\sharp(z, w) = \sum_{m \geq 1} c_m \Psi_m(z, w), \quad (10)$$

where

$$c_m = \frac{1}{m} \sum_{\substack{k|m \\ k \text{ odd}}} k \mu(k) = \frac{1}{m} \prod_{\substack{p|m \\ p \text{ odd}}} (1 - p). \quad (11)$$

For every m , $c_m \neq 0$ and $|c_m| \leq 1$. Consequently, for fixed $q > 0$,

$$\log \mathcal{Z}_{\text{orb}}(z, q) = \sum_{m \geq 1} c_m \frac{2(qz)^m}{1 - (1 + q^{2m})z^{2m}}. \quad (12)$$

Proof. Insert (7) into (8). A pair consisting of an odd Möbius index k and a repetition index r contributes

$$\frac{\mu(k)}{r} \frac{2w^{kr}}{1 - z^{2kr} - w^{2kr}}.$$

Grouping by $m = kr$ gives the divisor sum in (11). That sum factors over the odd radical of m , proving the product formula. No factor $1 - p$ vanishes. Moreover

$$|c_m| \leq \frac{1}{m} \prod_{\substack{p|m \\ p \text{ odd}}} p \leq 1.$$

Finally, $w = qz$ in (10) gives (12). $\square$

The coefficient c_m is independent of q : all weight dependence has moved into the geometry of the channel denominator. This separation is the main reason for using two fugacities rather than treating q as a passive real parameter.

4 The bidisk hypersurface divisor

Let $\mathbb{D}^2 = \{(z, w) \in \mathbb{C}^2 : |z| < 1, |w| < 1\}$ and define

$$\mathcal{H}_m = \{(z, w) \in \mathbb{D}^2 : z^{2m} + w^{2m} = 1\}, \quad \mathcal{H} = \bigcup_{m \geq 1} \mathcal{H}_m. \quad (13)$$

Theorem 4.1 (Smooth locally finite divisor). *Every $\mathcal{H}_m$ is a smooth reduced hypersurface. The formal sum $\sum_{m \geq 1} \mathcal{H}_m$ is a locally finite effective analytic divisor in the bidisk. The series in (10) converges normally on compact subsets of $\mathbb{D}^2 \setminus \mathcal{H}$ and therefore defines a holomorphic function $\mathcal{L}^\sharp$ there. Its exponential is a nonvanishing scalar continuation of the initial Euler germ.*

Proof. The gradient of $z^{2m} + w^{2m} - 1$ is

$$(2mz^{2m-1}, 2mw^{2m-1}).$$

It cannot vanish on $\mathcal{H}_m$, since simultaneous vanishing would imply $z = w = 0$. Thus $\mathcal{H}_m$ is smooth and reduced.

For a compact $K \subset \mathbb{D}^2$, choose $r, s < 1$ such that $|z| \leq r$ and $|w| \leq s$ on K . Once m is large enough, $r^{2m} + s^{2m} < 1$, so $K \cap \mathcal{H}_m$ is empty. This proves local finiteness.

Now suppose $K \subset \mathbb{D}^2 \setminus \mathcal{H}$. The finitely many early denominators in (9) have a positive minimum modulus on K . For all sufficiently large m ,

$$|1 - z^{2m} - w^{2m}| \geq 1 - r^{2m} - s^{2m} \geq \frac{1}{2}.$$

Together with $|c_m| \leq 1$, this gives $|c_m \Psi_m(z, w)| \leq 4s^m$. The Weierstrass test proves normal convergence. Agreement with the initial logarithm follows from theorem 3.1, and exponentiation gives the asserted continuation. $\square$

The theorem does not classify intersections $\mathcal{H}_m \cap \mathcal{H}_j$. Such joint complex collisions are unnecessary for the fixed-positive-weight fibers considered next.

5 Moving roots on positive-weight fibers

Fix $q > 0$ and restrict to $w = qz$. The m th denominator in (12) has the roots

$$\alpha_{m,\ell}(q) = \rho_m(q)e^{\pi i \ell/m}, \quad \rho_m(q) = (1 + q^{2m})^{-1/(2m)}, \quad 0 \leq \ell < 2m. \quad (14)$$

Proposition 5.1 (Strict radius separation). *For every $q > 0$, $\rho_m(q)$ is strictly increasing in m and*

$$\lim_{m \rightarrow \infty} \rho_m(q) = \min(1, q^{-1}). \quad (15)$$

Hence two different channels have no common root on a fixed positive- q fiber.

Proof. It suffices to show that $x \mapsto x^{-1} \log(1 + q^{2x})$ is strictly decreasing. Put $y = q^{2x}$. The numerator controlling its derivative is the negative of

$$(1 + y) \log(1 + y) - y \log y,$$

which is positive for $y > 0$. This proves strictness. The limit follows by separating $q < 1$, $q = 1$, and $q > 1$ in the defining formula. Distinct radii preclude a common complex root. $\square$

Theorem 5.2 (Exact fiberwise principal coefficient). *Let $q > 0$, $m \geq 1$, and $0 \leq \ell < 2m$. Near $\alpha = \alpha_{m,\ell}(q)$,*

$$\log \mathcal{Z}_{\text{orb}}(z, q) = \frac{c_m(-1)^\ell q^m}{m\sqrt{1 + q^{2m}}} \frac{1}{1 - z/\alpha} + G_{m,\ell,q}(z), \quad (16)$$

where $G_{m,\ell,q}$ is holomorphic. Thus the scalar continuation has an exponential essential singularity at every root in (14).

Proof. Set $v = 1 - z/\alpha$. Since $\alpha^{2m} = (1 + q^{2m})^{-1}$ and $\alpha^m = (-1)^\ell / \sqrt{1 + q^{2m}}$,

$$\begin{aligned} 1 - (1 + q^{2m})z^{2m} &= 2mv + O(v^2), \\ 2(qz)^m &= \frac{2(-1)^\ell q^m}{\sqrt{1 + q^{2m}}} + O(v). \end{aligned}$$

By proposition 5.1, no other channel denominator vanishes at α . Normal convergence from theorem 4.1 makes their sum holomorphic in a small neighborhood. Multiplication by the nonzero c_m gives (16); exponentiating a nonzero simple pole gives an essential singularity. $\square$

The radii and their limit locate the divisor but do not by themselves prove that complex roots approach every point of the limiting circle. We reserve that angular accumulation and the resulting natural-boundary question for a separate theorem.

6 Executable certificate and claim boundary

The primary certificate compares (12), coefficient by coefficient as exact polynomials in q , with the primitive/repetition law through degree 48. An independent sparse-polynomial implementation extends the comparison through degree 64 without importing the primary module. The test suite checks degree 100, reconstructs the first 128 channel coefficients, and audits all complex roots through channel 24 at $q = 1/2, 1, 2$. Dependency hashes lock the proofs, certificates, and PDFs of P69, P70, and P72. These finite checks guard the implementation; the all- m statements follow from the proofs above.

The resulting object has a genuine two-variable analytic divisor and exact fiberwise singular data. We make no comparison with a Lind zeta away from $q = 1$, because no such source formula has been supplied. We construct no transfer, nuclear, trace-class, or self-adjoint operator, and the integer channel label m has no certified rational-prime or von-Mangoldt meaning. No arithmetic advance or Route-B authorization follows.

References

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